

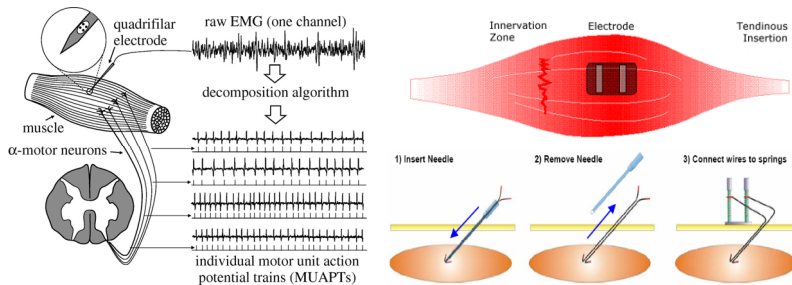
Electromyography Reusable Wearables for Targeted Muscle Rehabilitation and Growth

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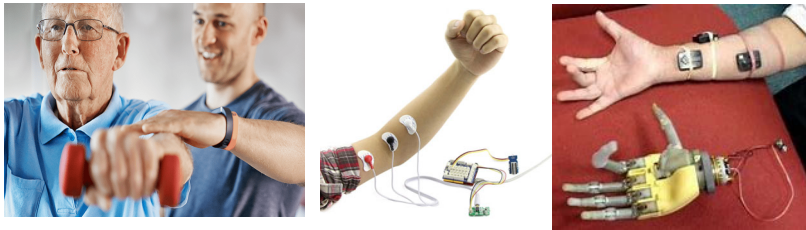
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Background

- Electromyography sensors (EMG) measures small electrical signals generated by your muscles during contractions. These muscle activations produce movements, such as; lifting your arm, clenching your fist, or moving a finger. Nerve signals from the brain, produce a burst of electrical activity (twitch signal) that circulates through adjoining bone and tissue and can be measured by electrodes on the skin.



- EMG sensors are sensitive enough to pick up the slightest movements of muscles. EMG systems have been gaining popularity in recent years for use in both athletics and physical rehabilitation. By tracking the muscular activity along with motions, wearable technology systems can provide valuable clinical data that can be crucial in the process of rehabilitation or exercise.

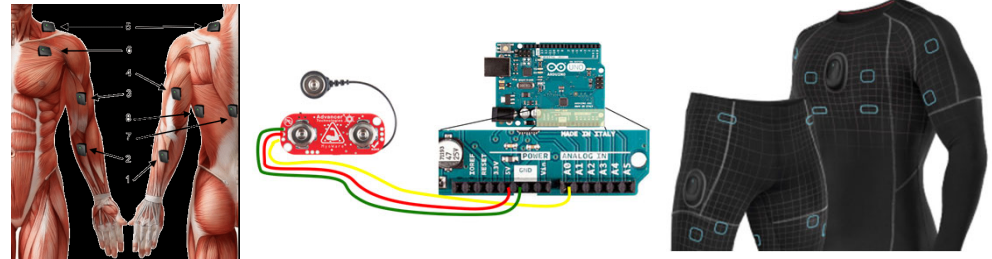


Objective

- The objective of the project is to use EMG sensors to target rehabilitation and create custom plans for people suffering from muscular degeneracy diseases or for injury rehabilitation.
- A wearable device could help improve therapy and shorten recovery time by targeting and tracking specific muscles in the affected area.

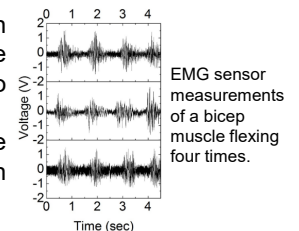
Materials and Methods

- EMG sensor placement relative to the muscle anatomy is important for data collection.
- Different open electronics, Arduino-based EMG sensors and filtering techniques were evaluated to help with assisted rehabilitation.
- A prototype wearable system with embedded EMG sensors will be created for physical therapists or patients who need muscle rehabilitation.
- The sensor data will be compared to existing consumer products, such as the Athos wearable fitness tracker.
- The cost and usability of the wearable systems will be improved to give patients real time feedback during exercise techniques and track their progress.



Results

- Testing for this system is ongoing. Preliminary studies have shown promising results for general muscle movements. However, more sensors are needed (the Myo system has 8 EMG sensors), to monitor muscles firing patterns for targeted treatment.
- Advanced digital filtering techniques are necessary to eliminate external and internal noise and to analyze signal information throughout multiple muscles in the target area.



Discussion

- Existing fitness suits with EMG sensors are extremely expensive and targeted for commercial use.
- They do not address some of the issues facing consumers, such as anatomy differences, maintaining sensor contact and reusability.
- Eventually, these wearable EMG sensors could be used by:
 - Fitness trainers
 - Rehabilitation
 - At home use